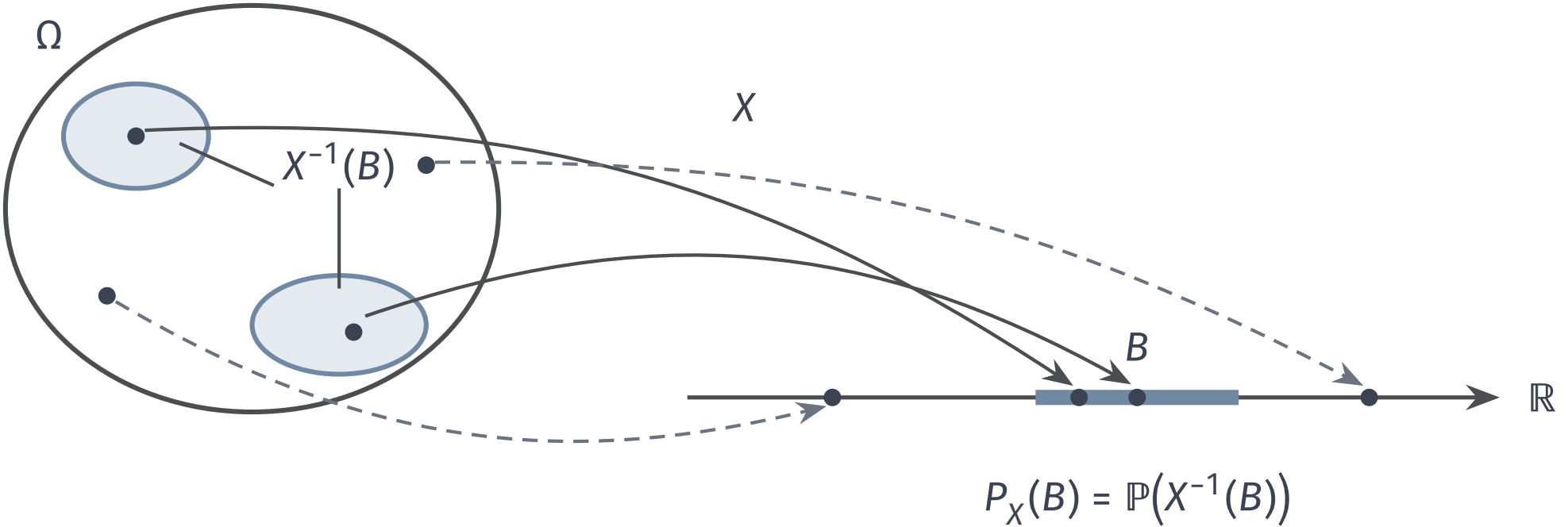
Ω
X
$X^{-1}(B)$
B
ℝ
$P_X(B) = \mathbb{P}\left(X^{-1}(B)\right)$